

Exercises

Exercise 8.1 Existence of representative consumer

Suppose households 1 and 2 have one-period utility functions $u(c^1)$ and $w(c^2)$, respectively, where u and w are both increasing, strictly concave, twice differentiable functions of a scalar consumption rate. Consider the Pareto problem:

$$v_\theta(c) = \max_{\{c^1, c^2\}} [\theta u(c^1) + (1 - \theta)w(c^2)]$$

subject to the constraint $c^1 + c^2 = c$. Show that the solution of this problem has the form of a concave utility function $v_\theta(c)$, which depends on the Pareto weight θ . Show that $v'_\theta(c) = \theta u'(c^1) = (1 - \theta)w'(c^2)$.

The function $v_\theta(c)$ is the utility function of the *representative consumer*. Such a representative consumer always lurks within a complete markets competitive equilibrium even with heterogeneous preferences. At a competitive equilibrium, the marginal utilities of the representative agent and each and every agent are proportional.

Exercise 8.2 Term structure of interest rates

Consider an economy with a single consumer. There is one good in the economy, which arrives in the form of an exogenous endowment obeying¹⁹

$$y_{t+1} = \lambda_{t+1}y_t,$$

where y_t is the endowment at time t and $\{\lambda_{t+1}\}$ is governed by a two-state Markov chain with transition matrix

$$P = \begin{bmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{bmatrix},$$

and initial distribution $\pi_\lambda = [\pi_0 \quad 1 - \pi_0]$. The value of λ_t is given by $\bar{\lambda}_1 = .98$ in state 1 and $\bar{\lambda}_2 = 1.03$ in state 2. Assume that the history of y_s, λ_s up to t is observed at time t . The consumer has endowment process $\{y_t\}$ and has preferences over consumption streams that are ordered by

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

¹⁹ Such a specification was made by Mehra and Prescott (1985).

where $\beta \in (0, 1)$ and $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$, where $\gamma \geq 1$.

a. Define a competitive equilibrium, being careful to name all of the objects of which it consists.

b. Tell how to compute a competitive equilibrium.

For the remainder of this problem, suppose that $p_{11} = .8$, $p_{22} = .85$, $\pi_0 = .5$, $\beta = .96$, and $\gamma = 2$. Suppose that the economy begins with $\lambda_0 = .98$ and $y_0 = 1$.

c. Compute the (unconditional) average growth rate of consumption, computed before having observed λ_0 .

d. Compute the time 0 prices of three risk-free discount bonds, in particular, those promising to pay one unit of time j consumption for $j = 0, 1, 2$, respectively.

e. Compute the time 0 prices of three bonds, in particular, ones promising to pay one unit of time j consumption contingent on $\lambda_j = \bar{\lambda}_1$ for $j = 0, 1, 2$, respectively.

f. Compute the time 0 prices of three bonds, in particular, ones promising to pay one unit of time j consumption contingent on $\lambda_j = \bar{\lambda}_2$ for $j = 0, 1, 2$, respectively.

g. Compare the prices that you computed in parts d, e, and f.

Exercise 8.3 An economy consists of two infinitely lived consumers named $i = 1, 2$. There is one nonstorable consumption good. Consumer i consumes c_t^i at time t . Consumer i ranks consumption streams by

$$\sum_{t=0}^{\infty} \beta^t u(c_t^i),$$

where $\beta \in (0, 1)$ and $u(c)$ is increasing, strictly concave, and twice continuously differentiable. Consumer 1 is endowed with a stream of the consumption good $y_t^1 = 1, 0, 0, 1, 0, 0, 1, \dots$. Consumer 2 is endowed with a stream of the consumption good $0, 1, 1, 0, 1, 1, 0, \dots$. Assume that there are complete markets with time 0 trading.

a. Define a competitive equilibrium.

- b.** Compute a competitive equilibrium.
- c.** Suppose that one of the consumers markets a derivative asset that promises to pay .05 units of consumption each period. What would the price of that asset be?

① We need to show that

$$v'_\theta(c) = \theta u'(c^1) = (1-\theta)w'(c^2)$$

and that v_θ is concave and depends on θ .

i) $v'_\theta(c) = \theta u'(c^1) = (1-\theta)w'(c^2)$

The problem is

$$v_\theta(c) = \max_{c^1, c^2} \{ \theta u(c^1) + (1-\theta)w(c^2) \}$$

$$\text{s.t. } c^1 + c^2 = c$$

Forming the Lagrangian

$$\mathcal{L} = \theta u(c^1) + (1-\theta)w(c^2) + \lambda(c - c^1 - c^2)$$

Taking the FOC's

$$\left. \begin{array}{l} c^1: \theta u'(c^1) - \lambda = 0 \\ c^2: (1-\theta)w'(c^2) - \lambda = 0 \end{array} \right\} \theta u'(c^1) = (1-\theta)w'(c^2) \quad (1)$$

since we can write c^2 as $c^2 = c - c^1$, we can write c^1 as endogenous to c ,

$$c^1 = (u')^{-1} \left(\frac{1-\theta}{\theta} w'(c - c^1) \right)$$

hence, $c^1(c)$. We can write the previous program as

$$v_\theta(c) = \theta u(c^1_\star(c)) + (1-\theta)w(c - c^1_\star(c))$$

where $c^1_\star(c)$ denotes the argmax of $v_\theta(c)$. Taking derivative wrt c ,

$$v'_\theta(c) = \theta u'(c^1_\star(c)) \frac{\partial c^1_\star(c)}{\partial c} + (1-\theta)w'(c - c^1_\star(c)) \left[1 - \frac{\partial c^1_\star(c)}{\partial c} \right] =$$

$$= \frac{\partial c^1_\star(c)}{\partial c} \left\{ \underbrace{\theta u'(c^1_\star(c)) - (1-\theta)w'(c - c^1_\star(c))}_{=0 \text{ by (1)!}} \right\} + (1-\theta)w'(c - c^1_\star(c))$$

Hence, we have proved it!

ii) v_θ concave and depends on θ

We know from the previous premise that

$$v'_\theta(c) = \theta u'(c^1(c))$$

taking the second derivative wrt c

$$v''_\theta(c) = \theta u''(c^1(c)) \frac{\partial c^1(c)}{\partial c}$$

⊖ since $u'(\cdot) > 0$ and $u''(\cdot) < 0$ (concavity) need to show ⊖

From (1) we know that

$$\theta u'(c^1(c)) = (1-\theta) w'(c - c^1(c))$$

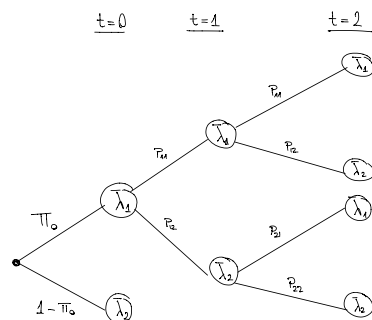
totally differentiating both sides wrt c

$$\theta u''(c^1(c)) \frac{dc^1(c)}{dc} = (1-\theta) w''(c - c^1(c)) \left[1 - \frac{dc^1(c)}{dc} \right] \Rightarrow$$

$$\Rightarrow \frac{dc^1(c)}{dc} = \frac{\overbrace{(1-\theta) w''(c - c^1(c))}^{\ominus}}{\underbrace{\theta u''(c^1(c))}_{\oplus} + \underbrace{(1-\theta) w''(c - c^1(c))}_{\ominus}} > 0 !$$

hence, we have proved that $v_\theta(c)$ is concave and depends on $\theta \in (0, 1)$!

②



a) Def: A competitive equilibrium is an initial endowment y_0 , an initial distribution π_λ , a consumption allocation $\{c_t(\lambda^t)\}_{t \in \mathbb{T}, \lambda^t}$ where $\lambda^t = \{\lambda_t\}_{t \in \mathbb{T}}$ is the entire history until time t , and a price system $\{q_t^0(\lambda^t)\}_{t \in \mathbb{T}, \lambda^t}$ such that

• Given y_0 and π_λ , the consumer solves

$$\begin{aligned} \max_{c_t(\tau)} \quad & \sum_{t=0}^{\infty} \sum_{\lambda^t} \beta^t \pi_t(\lambda^t) u(c_t(\lambda^t)) \\ \text{s.t.} \quad & \sum_{t=0}^{\infty} \sum_{\lambda^t} q_t^0(\lambda^t) c_t(\lambda^t) \leq \sum_{t=0}^{\infty} \sum_{\lambda^t} q_t^0(\lambda^t) y_t(\lambda^t) \end{aligned}$$

• Markets clear (feasibility): $c_t(\lambda^t) = y_t(\lambda^t) \quad \forall t$

b) The consumer problem is the previous one. Forming the Lagrangian

$$\mathcal{L} = \sum_{t=0}^{\infty} \sum_{\lambda^t} \left\{ \beta^t \pi_t(\lambda^t) \frac{c_t(\lambda^t)^{1-\gamma}}{1-\gamma} \right\} + \mu \left\{ \sum_{t=0}^{\infty} \sum_{\lambda^t} q_t^0(\lambda^t) [y_t(\lambda^t) - c_t(\lambda^t)] \right\}$$

Taking the FOC,

- i) only one representative agent
- ii) no time (time-0 trading)

$$c_t(\lambda^t): \beta^t \pi_t(\lambda^t) c_t(\lambda^t)^{-\gamma} - \mu q_t^0(\lambda^t) = 0$$

Notice that (i) there are no assets and (ii) goods are non-storable. Hence, at each time t , $c_t(\lambda^t) = y_t(\lambda^t)$ (given that $u(\cdot)$ is concave). Therefore, we can write price as

$$q_t^0(\lambda^t) = \frac{\beta^t \pi_t(\lambda^t) y_t(\lambda^t)^{-\gamma}}{\mu}$$

Setting $q_0^0(\lambda^0) = 1$ as numeraire, rewriting the FOC,

$$\underbrace{\beta^0 \pi_0(\lambda^0)}_1 c_0(\lambda^0)^{-\gamma} - \mu = 0 \Rightarrow \mu = c_0(\lambda^0)^{-\gamma}$$

Hence,

$$q_t^0(\lambda^t) = \beta^t \pi_t(\lambda^t) \left[\frac{y_t(\lambda^t)}{y_0(\lambda^0)} \right]^{-\gamma} \quad (2)$$

hence, both allocations and price are solved in terms of exogenous parameters.

c) Let us recap what we have. The transition matrix is

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.15 & 0.85 \end{pmatrix},$$

the initial distribution $\pi_\lambda = (0.5 \ 0.5)$ and the potential economy state $\lambda' = (0.98 \ 1.03)$. Let us now obtain the distribution in period 1.

$$\pi_1 = \pi_\lambda P = (0.5 \ 0.5) \begin{pmatrix} 0.8 & 0.2 \\ 0.15 & 0.85 \end{pmatrix} = \left(\frac{19}{40} \ \frac{21}{40} \right)$$

and consumption/endowment growth is given by

$$\mathbb{E}(g_c^1) = \pi_1 \lambda = \left(\frac{19}{40} \ \frac{21}{40} \right) \begin{pmatrix} 0.98 \\ 1.03 \end{pmatrix} = \frac{161}{160}$$

one could iterate this until infinity and obtain the stationary distribution π^* . However, since I do not derive utility from this, let us do it in the short way

$$\pi^* = \pi^* P \Rightarrow (\pi^* \ 1 - \pi^*) = (\pi^* \ 1 - \pi^*) \begin{pmatrix} 0.8 & 0.2 \\ 0.15 & 0.85 \end{pmatrix} \Rightarrow \pi^* = \frac{3}{7}; \pi^* = \left(\frac{3}{7} \ \frac{4}{7} \right)$$

and consumption/endowment growth is given by

$$\mathbb{E}(g_c^*) = \pi^* \lambda = \left(\frac{3}{7} \ \frac{4}{7} \right) \begin{pmatrix} 0.98 \\ 1.03 \end{pmatrix} = \frac{353}{350}$$

d) From (2) we know the price of an Arrow security that pays a unit of consumption at a given state-period.

More generally, the price of an asset that pays $d_t(\lambda^t)$ units of consumption at each time and history λ^t is

$$p_t^0(\lambda^t) = \sum_{t=0}^{\infty} \sum_{\lambda^t} q_t^0(\lambda^t) d_t(\lambda^t)$$

Hence, the price of a bond that pays 1 unit of consumption in every state ($d_t(\lambda^t)=1 \forall \lambda^t$), at a given period t is

$$Q_t^0(\lambda^t) = \sum_{\lambda^t} q_t^0(\lambda^t) = \sum_{\lambda^t} \beta^t \pi_t(\lambda^t) \left[\frac{y_t(\lambda^t)}{y_0(\lambda^0)} \right]^{-\gamma}$$

We knew from the text that $y_t = \lambda_t y_{t-1}$. Hence, we can write

$$\begin{aligned} y_1 &= \lambda_1 y_0 \\ y_2 &= \lambda_2 y_1 = \lambda_2 \lambda_1 y_0 \\ y_t &= \lambda_t \lambda_{t-1} \dots \lambda_1 y_0 \end{aligned}$$

Hence, we can write

$$Q_t^0(\lambda^t) = \sum_{\lambda^t} \beta^t \pi(\lambda_t | \lambda_{t-1}) \cdot \pi(\lambda_{t-1} | \lambda_{t-2}) \dots \pi(\lambda_2 | \lambda_1) \overbrace{\pi(\lambda_1 | \lambda_0)}^{\pi_{\bar{\lambda}_1}} \left[\frac{\lambda_t \lambda_{t-1} \dots \lambda_1 \cdot 1}{1} \right]^{-\gamma}$$

Therefore, the bond price at each t is

$$t=0$$

$$Q_0^0(\lambda^0) = \sum_{\lambda^0} \frac{\beta^0 \pi_0(\lambda^0) y_0(\lambda^0)^{-\gamma}}{y_0(\lambda^0)^{-\gamma}} = 1 \quad (\text{state } \bar{\lambda}_1 \text{ at } t=0 \text{ is certain, hence one unit of consumption costs, precisely, one unit of consumption})$$

$$t=1$$

$$Q_1^0(\lambda^1) = \sum_{\lambda^1} \frac{\beta^1 \pi_1(\lambda^1) y_1(\lambda^1)^{-\gamma}}{\underbrace{y_0(\lambda^0)^{-\gamma}}_1} = \beta [P_{11}(\bar{\lambda}_1 y_0)^{-\gamma} + P_{12}(\bar{\lambda}_2 y_0)^{-\gamma}] = 0.98$$

$$t=2$$

$$Q_2^0(\lambda^2) = \sum_{\lambda^2} \frac{\beta^2 \pi_2(\lambda^2) y_2(\lambda^2)^{-\gamma}}{\underbrace{y_0(\lambda^0)^{-\gamma}}_1} = \beta^2 [P_{11}^2(\bar{\lambda}_1^2 y_0)^{-\gamma} + P_{11}P_{12}(\bar{\lambda}_1 \bar{\lambda}_2 y_0)^{-\gamma} + P_{12}P_{22}(\bar{\lambda}_2^2 y_0)^{-\gamma} + P_{12}P_{21}(\bar{\lambda}_2 \bar{\lambda}_1 y_0)^{-\gamma}] = 0.95$$

e) Let me define $\bar{\lambda}_1 \equiv$ set of λ_1 states, and $\bar{\lambda}_2$ likewise. Hence, the price of a bond that pays only when $\bar{\lambda}_1$ is

$$Q_t^0(\lambda^t) = \sum_{\lambda^t \in \bar{\lambda}_1} \beta^t \pi_t(\lambda^t) \frac{y_t(\lambda^t)^{-\gamma}}{y_0(\lambda^0)^{-\gamma}}$$

$$t=0$$

$$Q_0^0(\lambda^0) = \sum_{\lambda^0 \in \bar{\lambda}_1} \frac{\beta^0 \pi_0(\lambda^0) y_0(\lambda^0)^{-\gamma}}{y_0(\lambda^0)^{-\gamma}} = 1 \quad (\text{certain } \bar{\lambda}_1 \text{ will happen})$$

$$t=1$$

$$Q_1^0(\lambda^1) = \sum_{\lambda^1 \in \bar{\lambda}_1} \frac{\beta^1 \pi_1(\lambda^1) y_1(\lambda^1)^{-\gamma}}{\underbrace{y_0(\lambda^0)^{-\gamma}}_1} = \beta [P_{11}(\bar{\lambda}_1 y_0)^{-\gamma}] = \frac{1920}{2401} \approx 0.80$$

$$t=2$$

$${}_1Q_2^0(\lambda^2) = \sum_{\lambda^2 \in \bar{\Lambda}_1} \frac{\beta^2 \pi_2(\lambda^2) y_2(\lambda^2)^{-r}}{\underbrace{y_0(\lambda^0)^{-r}}_1} = \beta^2 \left[P_{11}^2 (\bar{\lambda}_1^2 y_0)^{-r} + P_{12} P_{21} (\bar{\lambda}_2 \bar{\lambda}_1 y_0)^{-r} \right] = 0.67$$

f) The price of a bond that pays only when $\bar{\lambda}_2$ is

$${}_2Q_t^0(\lambda^t) = \sum_{\lambda^t \in \bar{\Lambda}_2} \beta^t \pi_t(\lambda^t) \frac{y_t(\lambda^t)^{-r}}{y_0(\lambda^0)^{-r}}$$

t=0

$${}_2Q_0^0(\lambda^0) = \sum_{\lambda^0 \in \bar{\Lambda}_2} \frac{\beta^0 \pi_0(\lambda^0) y_0(\lambda^0)^{-r}}{y_0(\lambda^0)^{-r}} = 0$$

(such state didn't happen! who would buy a bond that only pays in a state that isn't gonna happen)

t=1

$${}_2Q_1^0(\lambda^1) = \sum_{\lambda^1 \in \bar{\Lambda}_2} \frac{\beta^1 \pi_1(\lambda^1) y_1(\lambda^1)^{-r}}{\underbrace{y_0(\lambda^0)^{-r}}_1} = \beta \left[P_{12} (\bar{\lambda}_2 y_0)^{-r} \right] = \frac{1920}{10609} \approx 0.18$$

t=2

$${}_2Q_2^0(\lambda^2) = \sum_{\lambda^2 \in \bar{\Lambda}_2} \frac{\beta^2 \pi_2(\lambda^2) y_2(\lambda^2)^{-r}}{\underbrace{y_0(\lambda^0)^{-r}}_1} = \beta^2 \left[P_{11} P_{12} (\bar{\lambda}_1 \bar{\lambda}_2 y_0)^{-r} + P_{12} P_{22} (\bar{\lambda}_2^2 y_0)^{-r} \right] = 0.21$$

g) Notice that ${}_1Q_t^0(\lambda^t)$ is decreasing in time, and ${}_2Q_t^0(\lambda^t)$ is the reverse! As time goes by, the probability of being at $\bar{\lambda}_2$ increases (we saw it in part (c)). Hence, the seller of such a security should take that into account!

③ a) Def: A competitive equilibrium is a feasible allocation $\{c_t^i\}_{t,i}$ and a price system $\{q_t^o\}_t$ such that:

- Given the price system, the allocation solves each agent's problem

$$\begin{aligned} \max_{c_t^i} \quad & \sum_{t=0}^{\infty} \beta^t u(c_t^i) \\ \text{s.t.} \quad & \sum_{t=0}^{\infty} q_t^o c_t^i \leq \sum_{t=0}^{\infty} q_t^o y_t^i \end{aligned}$$

- Market clearing (feasibility): $\sum_{i=1}^{I=2} c_t^i = \sum_{i=1}^{I=2} y_t^i \quad \forall t$

b) Forming the Lagrangian of the previous problem

$$\mathcal{L} = \sum_{t=0}^{\infty} \left\{ \beta^t u(c_t^i) \right\} + \mu_i \left\{ \sum_{t=0}^{\infty} q_t^o [y_t^i - c_t^i] \right\}$$

Taking the FOC,

$$c_t^i: \beta^t u'(c_t^i) - \mu_i q_t^o = 0$$

(3)

Since prices q_t^o are the same for $i=1,2$,

$$q_t^0 = q_t^0 \Rightarrow \cancel{\beta^t} \frac{u'(c_t^1)}{\mu_1} = \cancel{\beta^t} \frac{u'(c_t^2)}{\mu_2} \Rightarrow c_t^1 = (u')^{-1} \left[\frac{\mu_1}{\mu_2} u'(c_t^2) \right]$$

From market clearing (feasibility)

$$\sum_{i=1}^{I=2} c_t^i = \sum_{i=1}^{I=2} y_t^i \Rightarrow c_t^1 + c_t^2 = \underbrace{y_t^1 + y_t^2}_{=1 \forall t} \Rightarrow (u')^{-1} \left[\frac{\mu_1}{\mu_2} u'(c_t^2) \right] + c_t^2 = 1$$

$$\left[\text{using } u(c_t^i) = \frac{(c_t^i)^{1-\gamma}}{1-\gamma} ; c_t^i = \frac{1}{1 + \left(\frac{\mu_1}{\mu_i} \right)^{-1/\gamma}} \equiv c^i \right]$$

Notice that individual consumption depends only on shadow price and aggregate endowment (which is constant!). Hence, consumption will be constant

$$c_t^1 = \bar{c}^1 ; c_t^2 = \bar{c}^2 \quad \forall t$$

Let us come back to (3). We can write price as

$$q_t^0 = \beta^t \frac{u'(c_t^i)}{\mu_i} \quad (4)$$

Set $q_0^0 = 1$ as numeraire. Rewriting (3) at $t=0$,

$$\beta^0 u'(c_0^i) = \mu_i$$

Henceforth we can rewrite (4) as

$$q_t^0 = \beta^t \frac{u'(c_t^i)}{u'(c_0^i)}$$

We just proved that consumption is constant! The previous expression then collapses to $q_t^0 = \beta^t$.

Each consumer budget constraint reads

$$\underbrace{\sum_{t=0}^{\infty} \beta^t \bar{c}^i}_{\frac{\bar{c}^1}{1-\beta} ; \frac{\bar{c}^2}{1-\beta}} = \underbrace{\sum_{t=0}^{\infty} \beta^t y_t^i}_{\frac{1}{1-\beta^3} ; \frac{\beta+\beta^2}{1-\beta^3}} \Rightarrow \begin{matrix} \bar{c}^1 = \frac{1-\beta}{1-\beta^3} \\ \bar{c}^2 = \frac{\beta-\beta^3}{1-\beta^3} \end{matrix} \quad \left(\text{it can be shown that } \bar{c}^1 > \bar{c}^2 \text{ only when } \beta > 2, \text{ which can never be the case in this setting} \right)$$

Hence, individual 2 consumes more (makes sense since he receives $\frac{2}{3}$ of the endowment stream).

c) We have already seen that $q_t^0 = \beta^t$, which is the price of Arrow securities. The price of such an asset would be

$$P_t^0 = \sum_{t=0}^{\infty} q_t^0 (\Delta \bar{c}^i) = \sum_{t=0}^{\infty} \beta^t 0.05 \Rightarrow P_t^0 = \frac{0.05}{1-\beta}$$